I-ADIC COMPLETENESS OF A MODULE IS EQUIVALENT TO THE VANISHING OF ITS TORSION FUNCTORS AND SURJECTIVITY ONTO ITS COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 25962-25970 (canonical tag 091R) and the counted proof is the proof environment at source lines 25972-26015, which immediately follows it and is its own proof.

Editorial note. The proof refers to lemma-hom-from-Af inside this chapter and to algebra-lemma-when-surjective-to-completion of another chapter; both are navigation pointers with their roles recorded in the external reference reasons. The infinite correction series of part (1) and the universal extension of part (2) are inside the counted span.

Editorial note. The proof is divided by the author into the two numbered parts with medskip headings, and both are delivered in full.

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Allowed standard prerequisites (editorial).

- For an element f of a ring, the torsion functor T(-, f) is computed by the Hom and Ext groups from the localization A_f, so its vanishing is equivalent to the vanishing of Hom_A(A_f, M) and Ext^1_A(A_f, M). invoked at proof lines 25974-25975 and again at proof lines 26005-26006. Turns the vanishing condition on the torsion functor into two explicit vanishing statements about maps out of A_f, which are then proved by hand.
- An I-adically complete module is the inverse limit of its quotients M over I^n, and Hom from A_f into a module annihilated by a power of f vanishes. used at proof lines 25976-25977. Gives the vanishing of Hom_A(A_f, M) for a complete module.
- The snake lemma identifies the quotients M over f^n M and E over f^n E for an extension of M by A_f, and a short exact sequence of modules in which the outer terms are killed by a power of f splits. used at proof

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lines 26012-26013 and at proof lines 26005-26007. Produces the congruence expressing a given element x as a finite combination of the x_n modulo f^n M, which is the surjectivity assertion of part (2).

Source reference. Original lemma 091P: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25850–25850. The reference identifies the vanishing of T(M, f) with the vanishing of Hom and Ext from A_f, used at proof lines 25974-25975 and 26005-26006; the two explicit constructions that verify those vanishing conditions are inside the counted span.

Source reference. Original lemma 090S: sources/stacks-chow/tags.txt, source lines 11420–11420. The reference is the reduction step that lets the finitely generated case be replaced by the case of one element, used at proof lines 25993-25995; the extension built afterwards and the snake lemma argument are inside the counted span.

Lemma 0.2. *Let A be a ring. Let $I \subset A$ be an ideal. Let M be an A-module.*

(1) *If M is I-adically complete, then $T(M, f) = 0$ for all $f \in I$.*
(2) *Conversely, if $T(M, f) = 0$ for all $f \in I$ and I is finitely generated, then $M \to \lim M/I^n M$ is surjective.*

Proof. Proof of (1). Assume M is I-adically complete. By Lemma 091P it suffices to prove $\mathrm{Ext}^1_A(A_f, M) = 0$ and $\mathrm{Hom}_A(A_f, M) = 0$. Since $M = \lim M/I^n M$ and since $\mathrm{Hom}_A(A_f, M/I^n M) = 0$ it follows that $\mathrm{Hom}_A(A_f, M) = 0$. Suppose we have an extension

$$0 \to M \to E \to A_f \to 0$$

For $n \geq 0$ pick $e_n \in E$ mapping to $1/f^n$. Set $\delta_n = fe_{n+1} - e_n \in M$ for $n \geq 0$. Replace e_n by

$$e'_n = e_n + \delta_n + f\delta_{n+1} + f^2\delta_{n+2} + \ldots$$

The infinite sum exists as M is complete with respect to I and $f \in I$. A simple calculation shows that $fe'_{n+1} = e'_n$. Thus we get a splitting of the extension by mapping $1/f^n$ to e'_n.

Proof of (2). Assume that $I = (f_1, \ldots, f_r)$ and that $T(M, f_i) = 0$ for $i = 1, \ldots, r$. By Algebra, Lemma 090S we may assume $I = (f)$ and $T(M, f) = 0$. Let $x_n \in M$ for $n \geq 0$. Consider the extension

$$0 \to M \to E \to A_f \to 0$$

given by

$$E = M \oplus \bigoplus Ae_n \Big/ \langle x_n - fe_{n+1} + e_n \rangle$$

mapping e_n to $1/f^n$ in A_f (see above). By assumption and Lemma 091P this extension is split, hence we obtain an element $x + e_0$ which generates a copy of A_f in E. Then

$$x + e_0 = x - x_0 + fe_1 = x - x_0 - fx_1 + f^2 e_2 = \ldots$$

Since $M/f^n M = E/f^n E$ by the snake lemma, we see that $x = x_0 + fx_1 + \ldots + f^{n-1}x_{n-1}$ modulo $f^n M$. In other words, the map $M \to \lim M/f^n M$ is surjective as desired. □